

Data Processing

Chapter 2 – Hands on Data Analytics for Everyone

September 20, 2026

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Basic Descriptive Stats: Mean and Variance



Statistical measures can be used to describe a data field

Given X = [78, 85, 92, 85, 70, 88, 95, 85, 90, 62]

- Max(X) 95
- Min(X) 62
- Range(X) = Max(X) – Min(X) 33

- Mean: $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$

  (78+85+92+85+70+88+95+85+90+62)/10 = 83

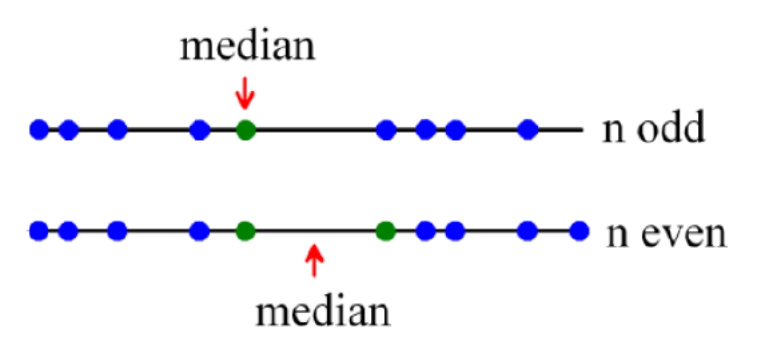
- Variance $\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\mu)^2$

  $[(78-83)^2+(85-83)^2+ \ldots..+ (62-83)^2]$ /9 ≈ **105.56**

- **Standard Deviation =** $\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\mu)^2}$

  $\sigma = \sqrt{105.56}$ ≈ **10.27**

Basic Descriptive Stats: Quantiles



- **Median:** The value in the middle (for values sorted in increasing order)
- **q%-quantile** (0 < q < 100): The value for which q% of the values are smaller and 100-q% are larger. The median is the 50%-quantile
- **Quartiles**: 25%-quantile (1st quartile), median (2nd quantile), 75%-quantile (3rd quartile)
- **Interquartile range (IQR):** 3rd quartile – 1st quartile

How to find quartiles



- Count the number of observations in the dataset (*n*).
- Sort the observations from smallest to largest.
- Find the first quartile:
  - Calculate *n**(1/4).
  - If *n**(1/4) is an integer, then the first quartile is the mean of the numbers at positions *n**(1/4) and *n**(1/4) + 1.
  - If *n**(1/4) is **not** an integer, then round it **up**. The number at this position is the first quartile.

How to find quartiles



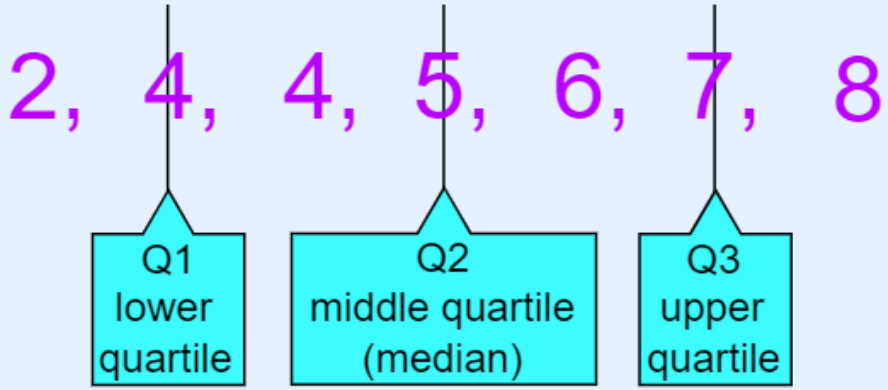
- Find the second quartile:
  - Calculate $n*(2/4)$.
  - If $n*(2/4)$ is an integer, the second quartile is the mean of the numbers at positions $n*(2/4)$ and $n*(2/4) + 1$.
  - If $n*(2/4)$ is **not** an integer, then round it **up**. The number at this position is the second quartile.
- Find the third quartile:
  - Calculate $n*(3/4)$.
  - If $n*(3/4)$ is an integer, then the third quartile is the mean of the numbers at positions $n*(3/4)$ and $n*(3/4)+1$.
  - If $n*(3/4)$ is **not** an integer, then round it **up**. The number at this position is the third quartile.

Quartiles - Example

Example: 5, 7, 4, 4, 6, 2, 8

Put them in order: 2, 4, 4, 5, 6, 7, 8

Cut the list into quarters:



And the result is:

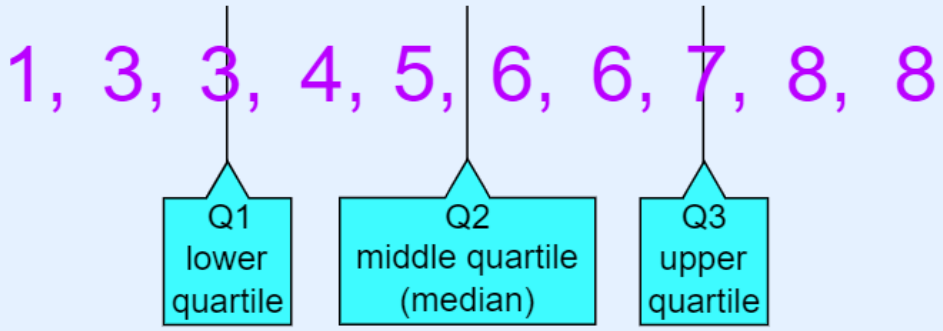
- Quartile 1 (Q1) = **4**
- Quartile 2 (Q2), which is also the Median, = **5**
- Quartile 3 (Q3) = **7**

Quartiles - Example

Example: 1, 3, 3, 4, 5, 6, 6, 7, 8, 8

The numbers are already in order

Cut the list into quarters:



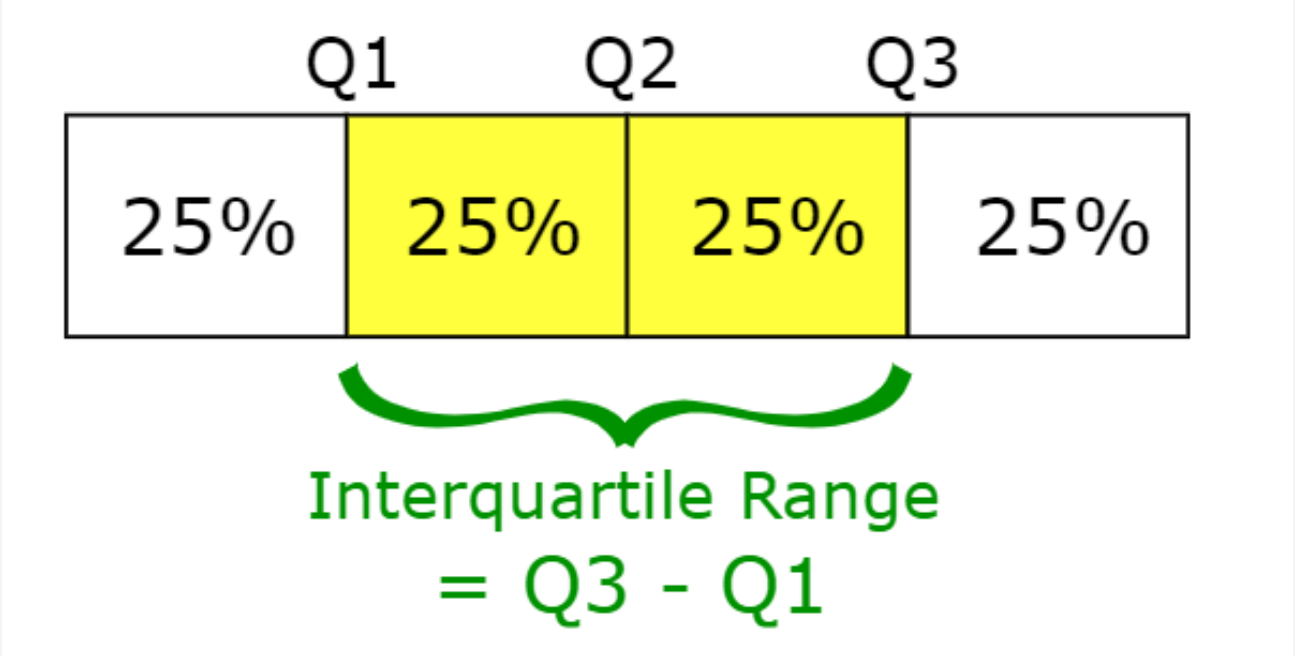
In this case Quartile 2 is half way between 5 and 6:

$$Q2 = (5+6)/2 = \mathbf{5.5}$$

And the result is:

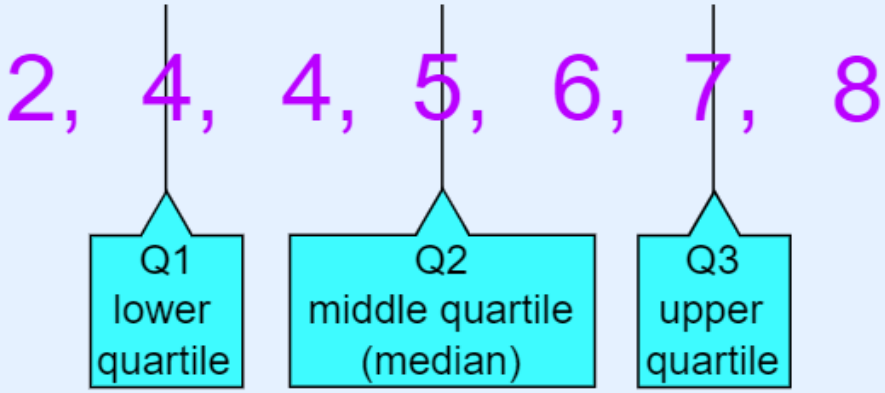
- Quartile 1 (Q1) = **3**
- Quartile 2 (Q2) = **5.5**
- Quartile 3 (Q3) = **7**

Quartiles - Example



Quartiles - Example

Example:



The **Interquartile Range** is:

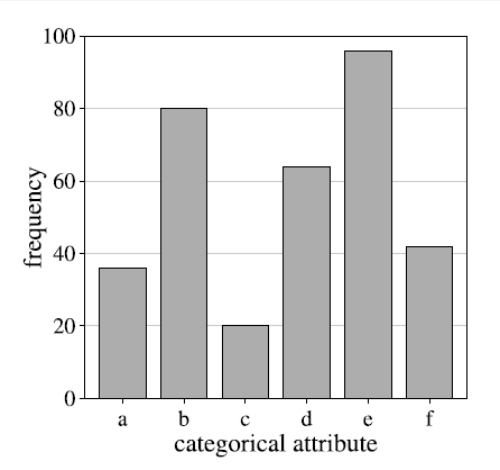
$$Q3 - Q1 = 7 - 4 = \mathbf{3}$$

https://www.mathsisfun.com/data/quartiles.html

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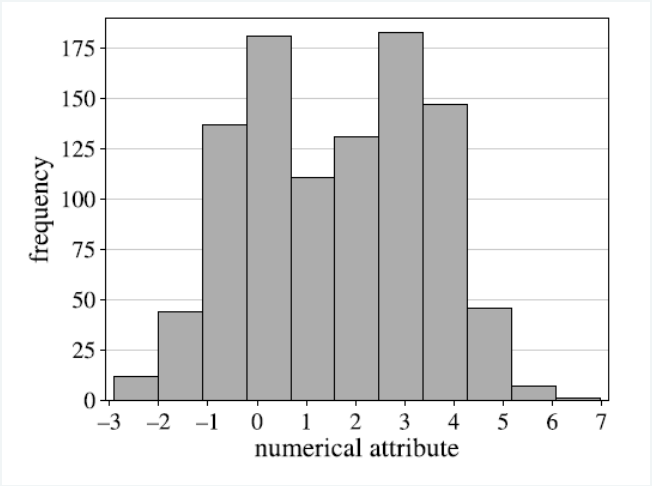
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Bar Chart



A bar chart is a simple way to depict the frequencies of the values of a categorical attribute.

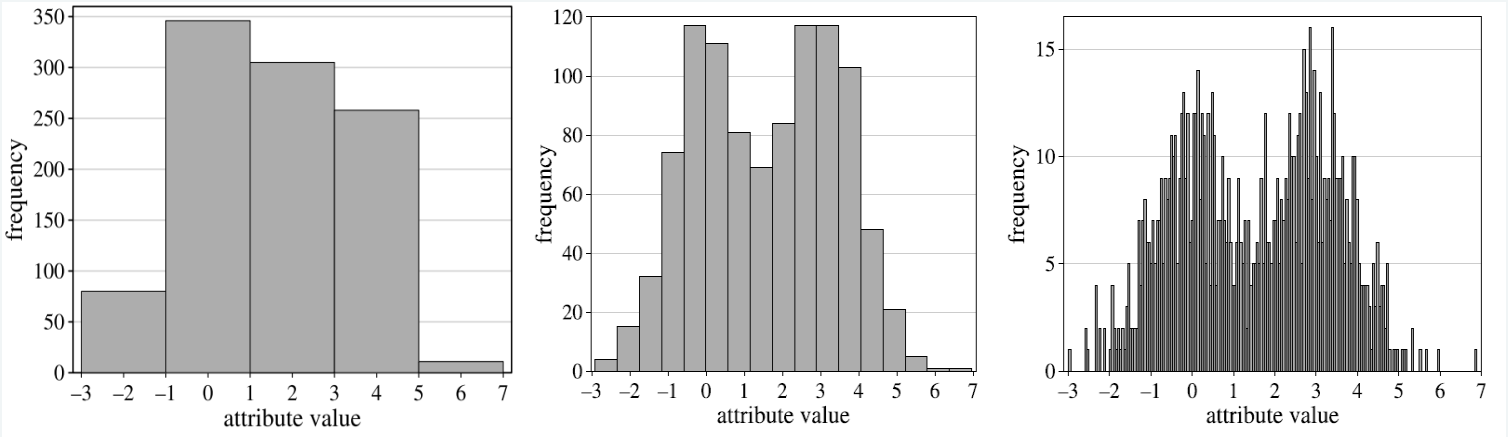
Histogram



- A histogram shows the frequency distribution for a numerical attribute.
- The range of the numerical attribute is discretized into a fixed number of intervals (bins), usually of equal length.
- For each interval, the (absolute) frequency of values falling into it is indicated by the height of a bar.

Choice of Number of Bins

The histogram in the figures resulted from a sample of size $n = 1000$



Choosing a low number of bins, the two peaks of the original distribution are no longer visible, and one gets the wrong impression that the distribution is unimodal

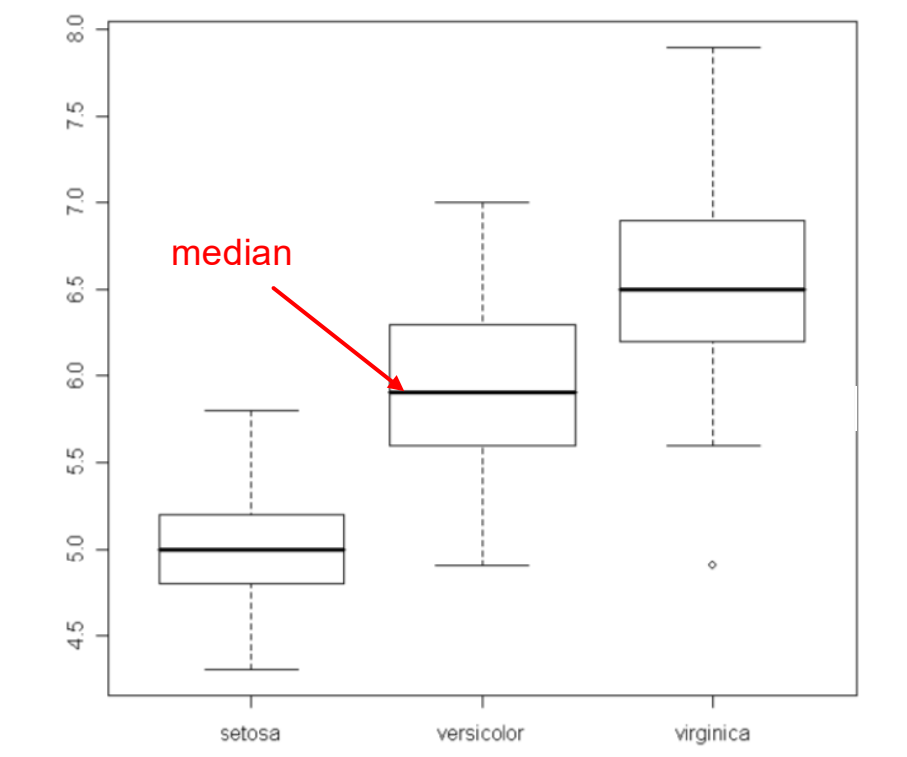
Choosing a high number of bins usually leads to a very scattered histogram in which it is difficult to distinguish true peaks from random peaks

Best choice for number k of bins in the histogram?

- Sturge's Rule $k = \lceil log_2(n) + 1 \rceil$

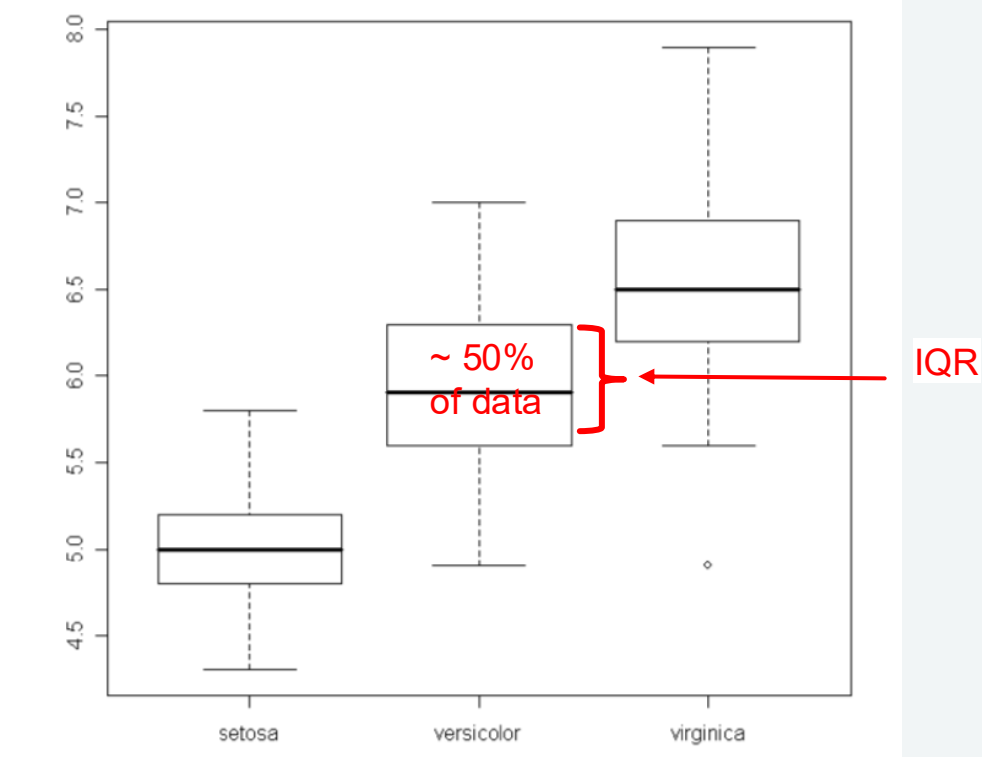
Boxplots

Boxplots are a very compact way to visualize and summarize the main characteristics of a numeric attribute, through the ***median***, the IQR, and possible outliers



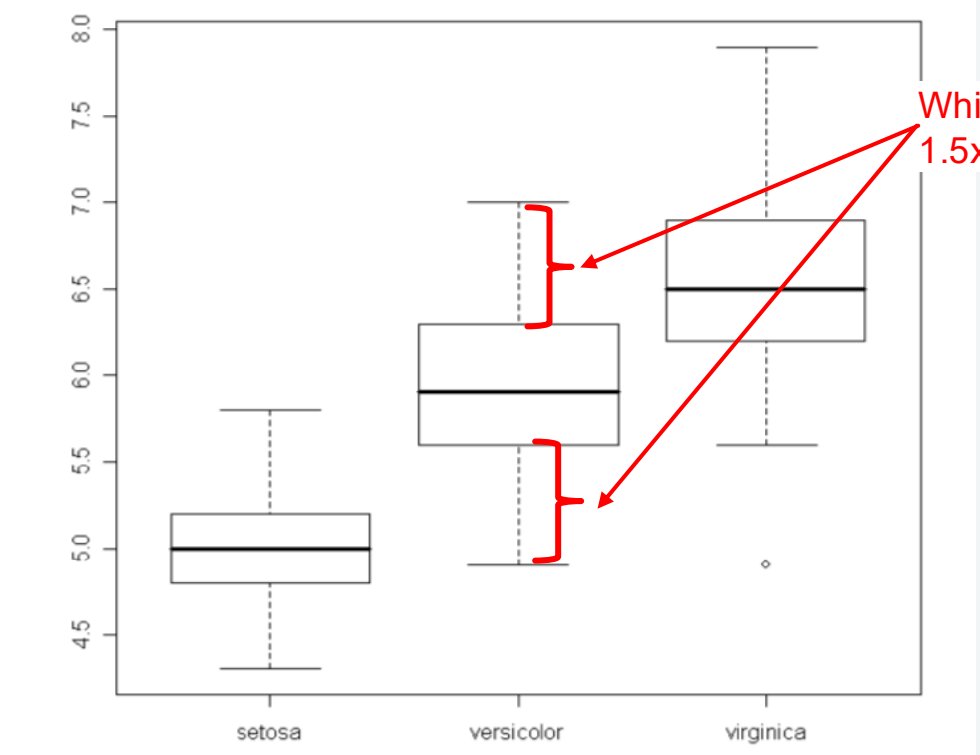
Boxplots

Boxplots are a very compact way to visualize and summarize the main characteristics of a numeric attribute, through the median, the ***IQR***, and possible outliers



Boxplots

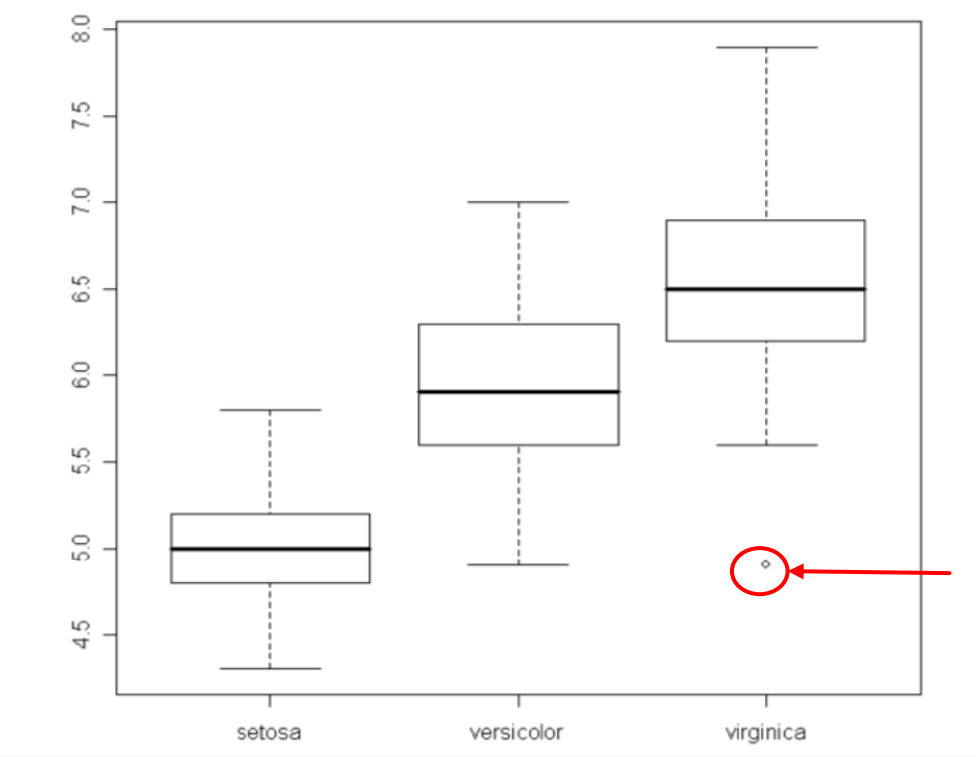
Boxplots are a very compact way to visualize and summarize the main characteristics of a numeric attribute, through the median, the ***IQR***, and possible outliers



The maximum length of each whisker is 1.5 times the length of the interquartile range. But if there is no data point at the maximum length of a whisker, the corresponding whisker is shortened until it reaches the next data point.

Boxplots

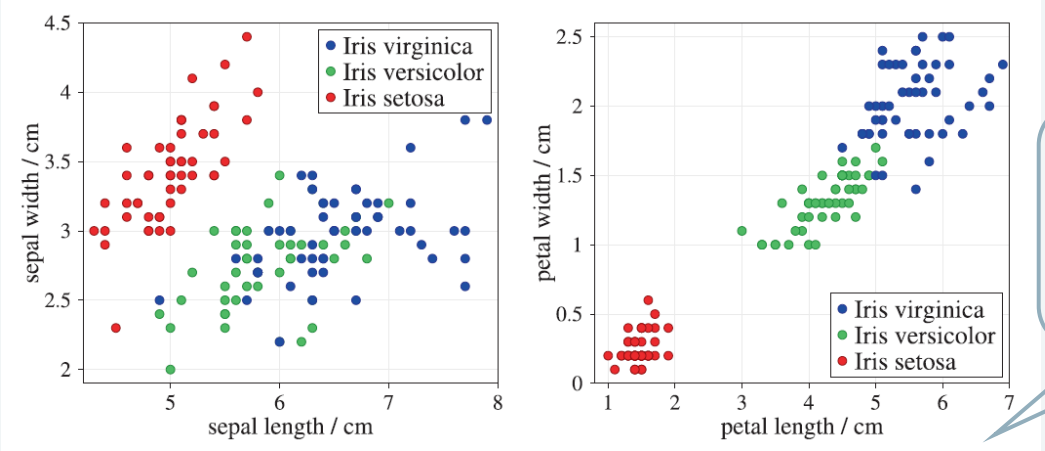
Boxplots are a very compact way to visualize and summarize the main characteristics of a numeric attribute, through the median, the *IQR*, and possible **outliers**



Data points lying outside the whiskers are considered as outliers and are indicated in the form of small circles.

Scatter Plot for Two Attributes



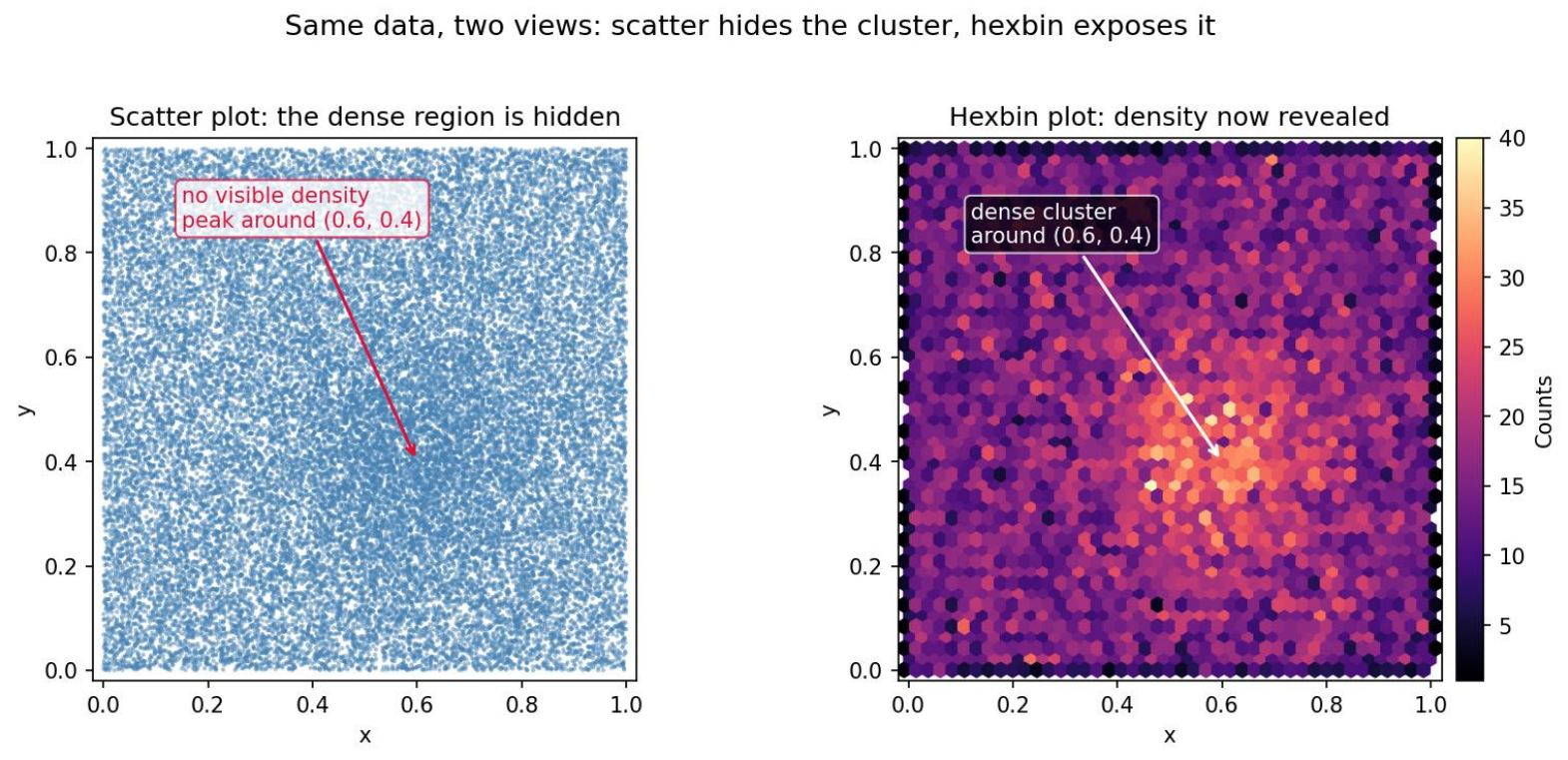


Petal length and width provide better class separation than sepal length and width

Scatter plots of the Iris(鸢(yuan)尾花) data set for sepal(萼片) length vs. sepalwidth (left) and for petal (花瓣) length vs. petal width (right). All quantities are measured in centimetres

- In scatter plots two attributes are plotted against each other
- Can be enriched with additional features (color, shape, size)
- Suitable for small number of points; not suitable for large datasets
- Points can hide each other -> add **Jitter** (a small random value to each point)

Scatter Plot for Two Attributes



Scatter plot is not suitable for large datasets. Both plots indicate a higher density of the data around the point (0.6, 0.4).

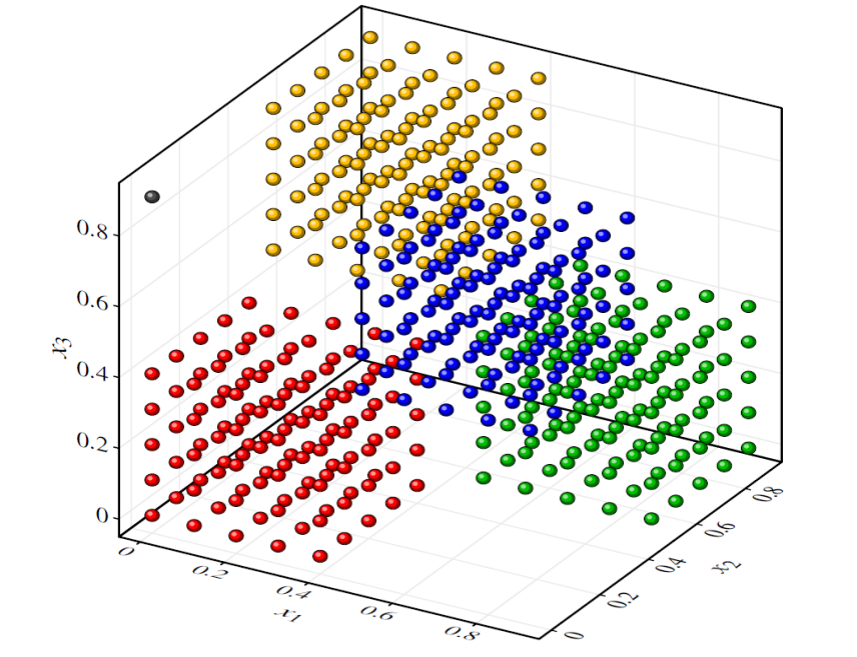
- Simple scatter plot cannot distinguish higher density
- Density plot (hexagonal binning) groups points in color (or semi-transparent) hexagons: the more points in the same place the stronger the color (or less transparent)

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Visualization of Three-dimensional Data: 3D Plot

A display or plot is **by definition two-dimensional**, so that only maximum two axes (attributes) can be incorporated. **3D** techniques can be used to incorporate three axes (attributes).



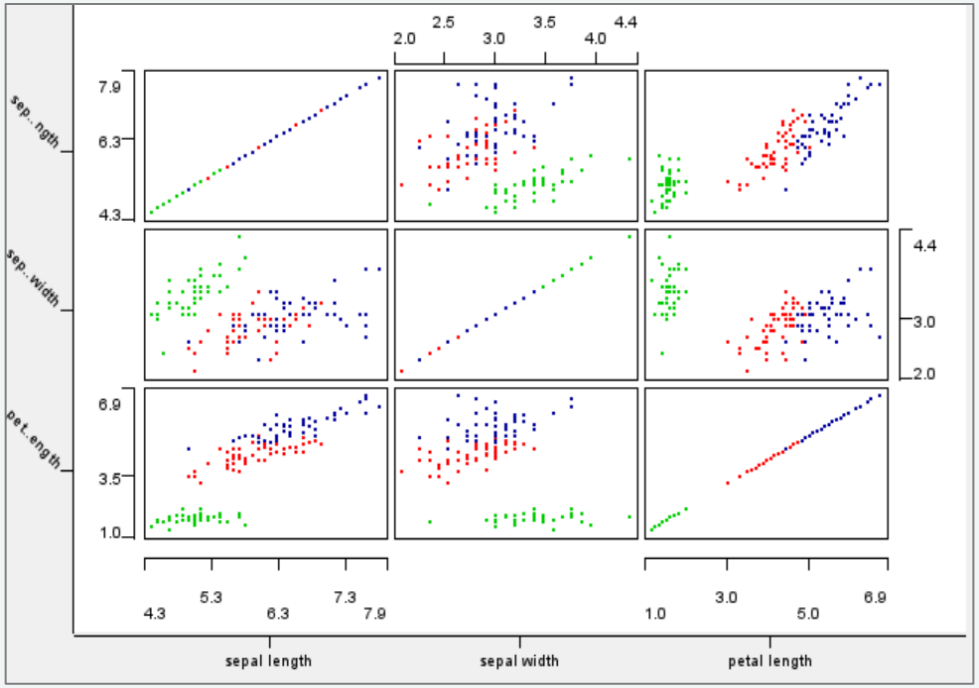
3D scatter plot

Example

- A data set distributed over a cube in a **chessboard-like pattern**.
- The colors are only meant to make the different cubes more easily discernible (可辨别的). They do not indicate classes.
- Note the outlier in the upper left corner

Visualization of Three-dimensional Data: Scatter Matrixes

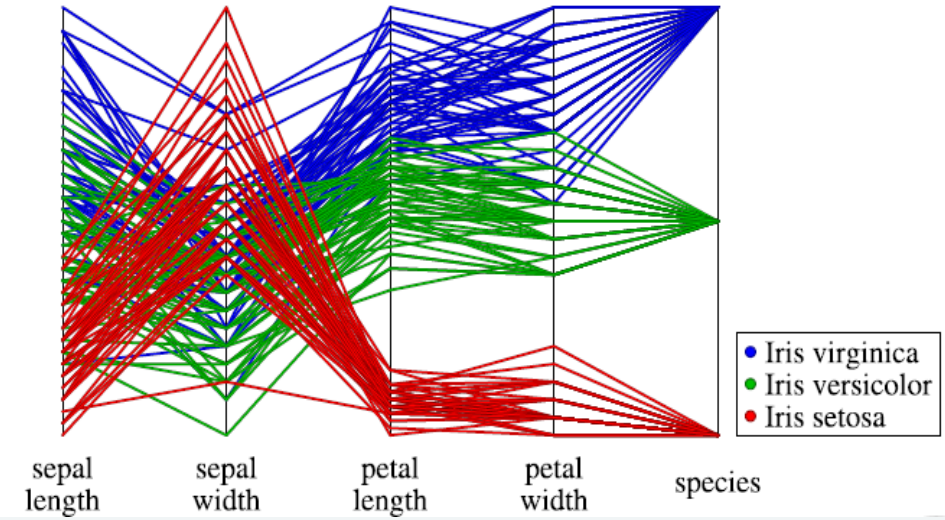
- A matrix of scatter plots $m \times m$ where m is the number of attributes (data dimensionality)
- For *m* attributes there are $\binom{m}{2} = m(m-1)/2$ possible scatter plots
- e.g. For 50 attributes there are 2450/2 scatter plots!



Scatter matrix

Parallel Coordinates Plot

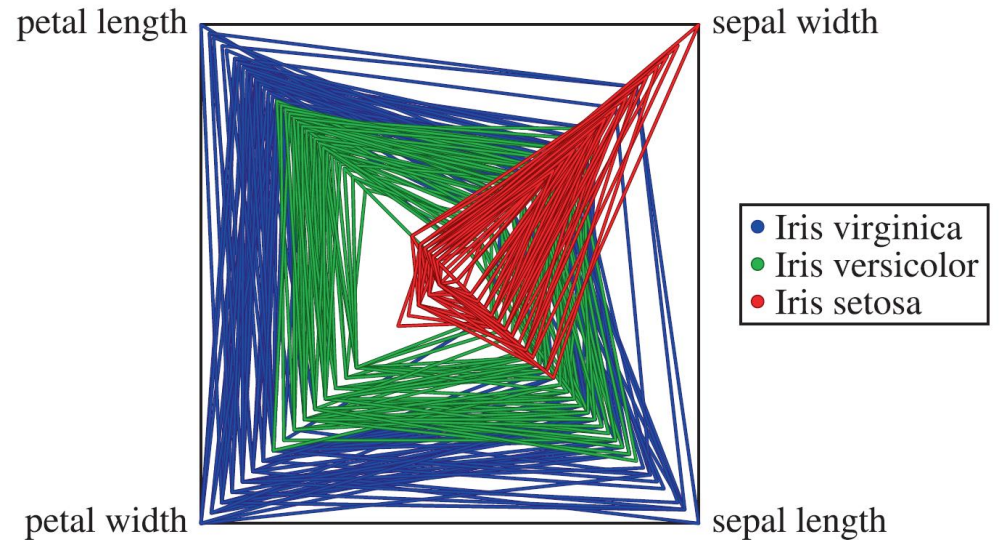
- Parallel coordinates draw the coordinate axes for each attribute parallel to each other, so that there is no limitation for the number of axes to be displayed.
- For each data object, a polyline is drawn connecting the values of the attributes on the corresponding axes.
- Maintains the original attributes
- Limited number of entries
- How do we spot correlation between features?



Parallel coordinates plot for the Iris data set

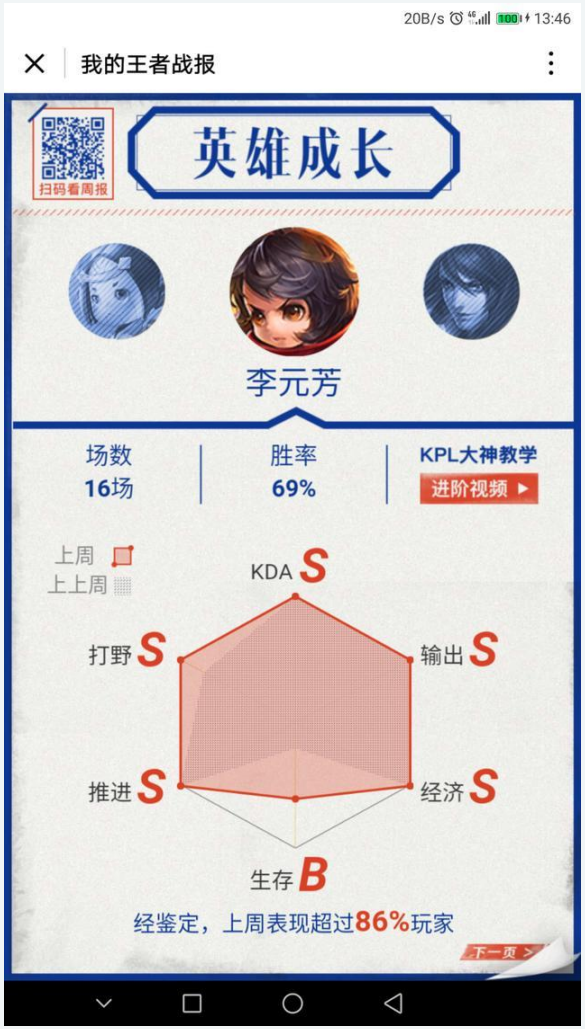
Radar Plot

- Similar idea of the Parallel Coordinates plot
- Axes are drawn in a star-like fashion intersecting in one point
- Also called spider plots
- Suitable for small datasets



Radar plot for the Iris data set

Radar Plot


20B/s 13:46
我的王者战报
英雄成长
扫码看周报
李元芳
场数
16场
胜率
69%
KPL大神教学
进阶视频
上周
上上周
KDA S
打野 S
输出 S
推进 S
经济 S
生存 B
经鉴定，上周表现超过86%玩家
下一页


当前赛季 (S14)
全部比赛
标准模式
输出
KDA
发育
团战
生存
场次
7
胜率
42. 9%
综合得分
74. 3
详细数据
分享
全场最佳2
败方最佳2
三连决胜0
四连超凡0
五连绝世0
超神0

Sunburst Chart

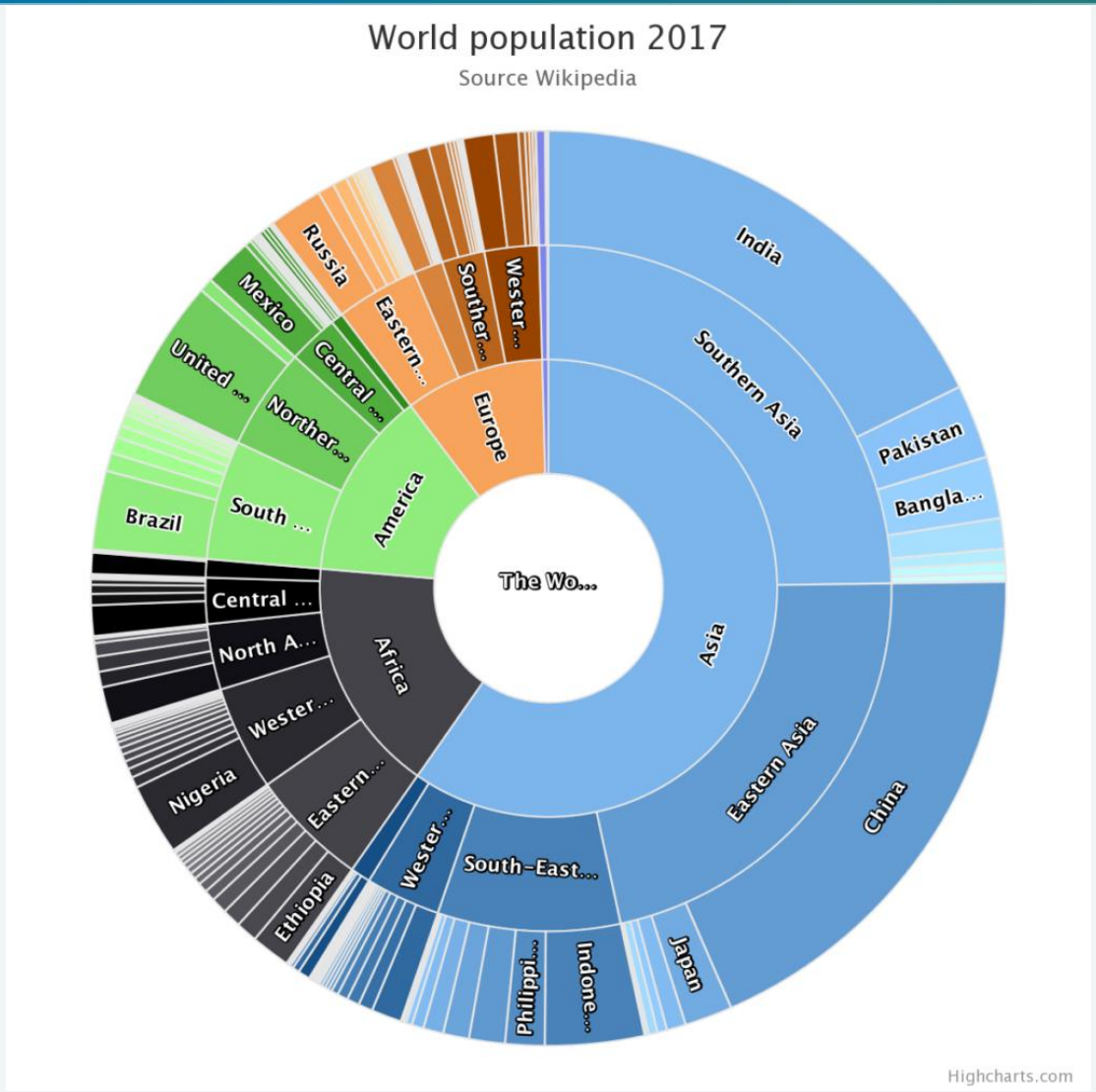


- Display multidimensional hierarchical nominal data in a radial layout
- One section ⇔ one attribute

e.g. Fruity → Berry → Blueberry

- Root attribute in the center, external sections are attributes located deeper in the hierarchy
- Area of a section represents the accumulated value of all descending sections

Sunburst Chart



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Missing Value Imputation: Motivation



Data is not always available

- E.g., many tuples have no recorded value for several attributes, such as weight in a people database

Missing data may be due to:

- Equipment malfunctioning (broken sensors)
- Inconsistency with other recorded data and thus deleted
- Data not entered (manually)
- Data not considered important at the time of collection
- Data format / contents of database changes
- Refusal to answer a question
- Irrelevant attribute for the corresponding object (pregnant (yes/no) for men)
- Missing value might not necessarily be indicated as missing (instead: zero or default values).

Missing Values: Types



Types of missing values:

Example: Suppose you are modeling weight Y as a function of sex X

- **Missing Completely At Random** (MCAR): the probability that a value for X is missing does neither depend on the value of X nor on other variables.
  There may be no particular reason why some people told you their weights and others didn't.

- **Missing At Random** (MAR): the probability that Y is missing depends only on the value of X.
  One sex X may be less likely to disclose its weight Y.

- **Not Missing At Random** (NMAR): the probability that Y is missing depends on the unobserved value of Y itself.
  Heavy (or light) people may be less likely to disclose their weight.

Missing Values Imputation



How to handle missing values?

- Ignore/delete the record
- Fill in (impute) missing value as:
  - **Fixed value**: e.g., “unknown”, -9999, -1 when only positive numbers in the domain, etc.
  - Attribute **mean / median**
  - Attribute most frequent value **(mode)**
  - **A predicted value** based on the other attributes (e.g. nearest neighbor approach)

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Outlier Detection



What are outliers?

- An **outlier** is a value or data object that is far away or very different from all or most of the other data.
- Outliers are supposed to be rare

Outlier Detection



Causes for outliers:

- Data **quality** problems (erroneous data coming from wrong measurements or typing mistakes)
- Exceptional or unusual **situations**/data objects.

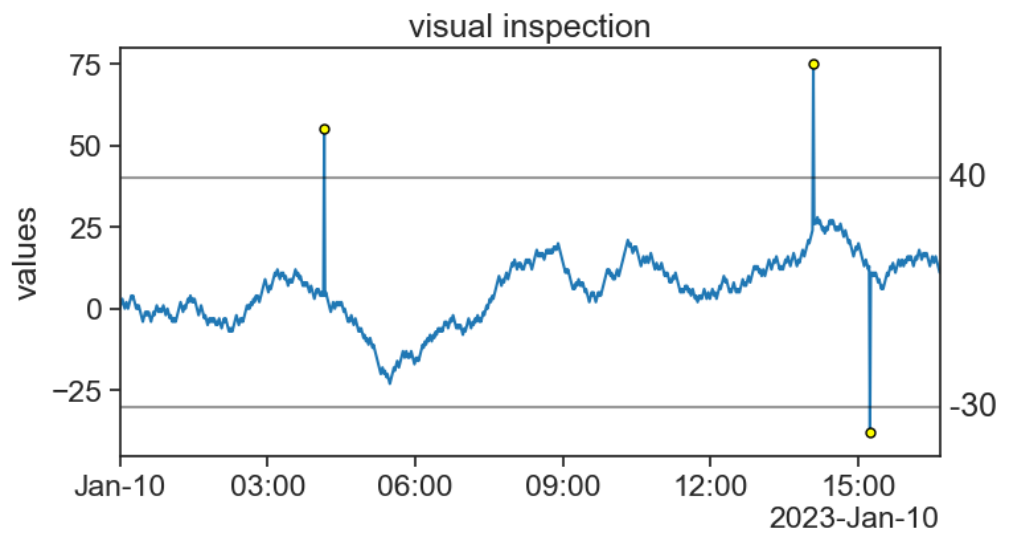
Outlier handling :

- Outliers coming from erroneous data should be **excluded** from the analysis.
- Even if the outliers are correct (exceptional data), it is sometimes useful to exclude them from the analysis.
- For example, a single extremely large outlier can lead to completely misleading values for the mean value.

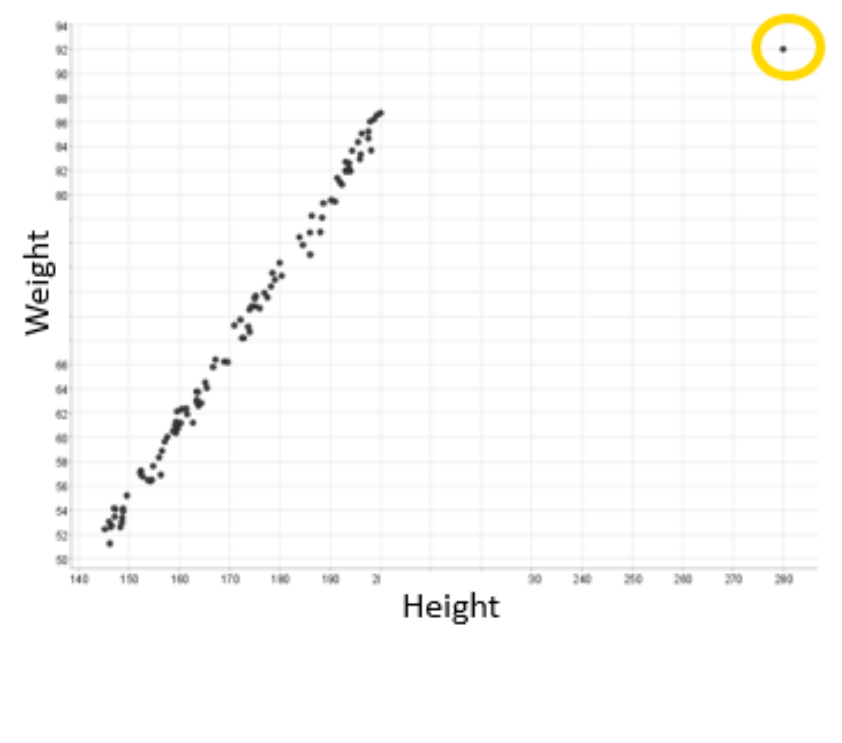
Outlier Detection

An outlier could be, for example, a rare behaviour, a system defect, a measurement error, or a reaction to an unexpected event

Outliers in time series (ordered data)

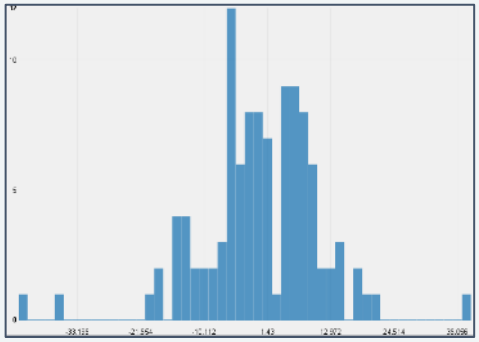


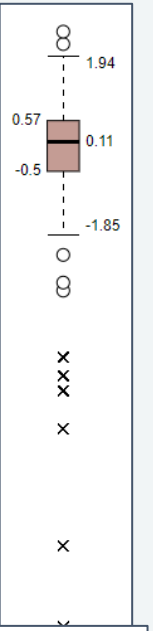
Outlier in 2 dimensions: (Height, Weight)

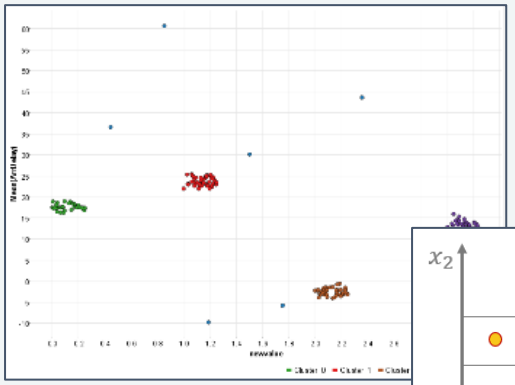


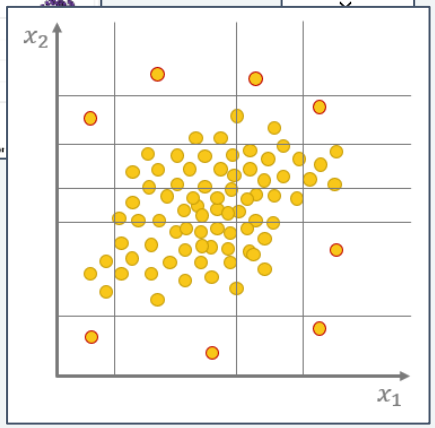
Outlier Detection Techniques

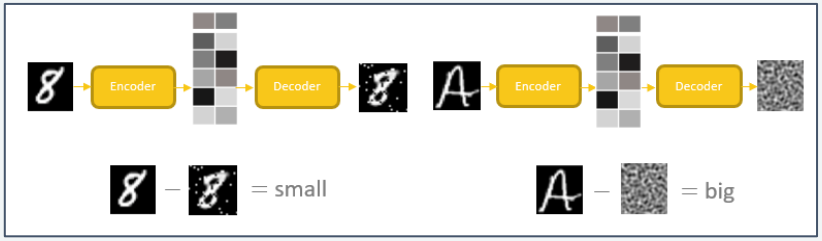
- Knowledge-based
  - We know that a 200 year old person must be a mistake
  - We know that "A" in a number corpus is an outlier
- Statistics-based
  - Distance from the median
  - Position in the distribution tails
  - Distance to the closest cluster center
  - Error produced by an autoencoder
  - Number of random splits to isolate a data point from other data











Statistical Methods for Outlier Detection

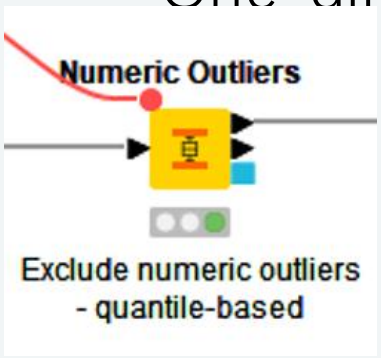


- Quantile-based: **Box plot**
- Distribution-based: **Z-Score**
- Cluster-based: **DBSCAN**
- Neural Autoencoder
- Isolation Forest
- …

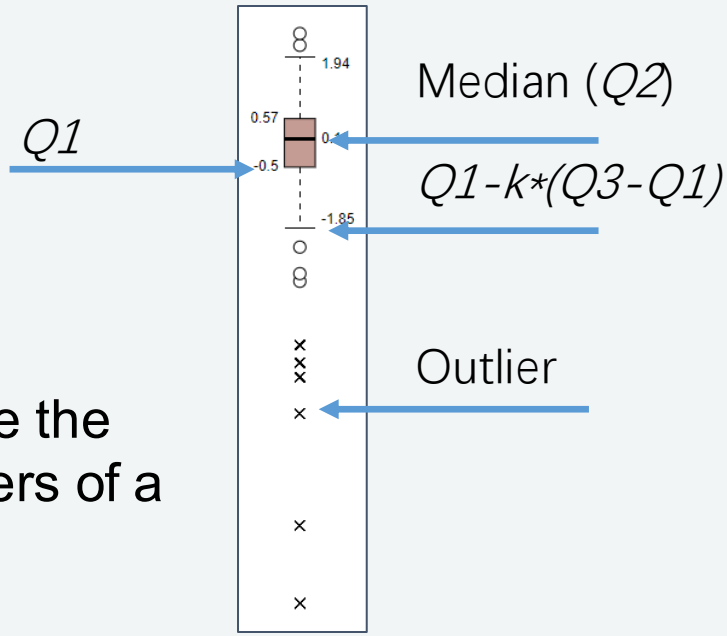
Quartile based (Box Plots)

Challenges:

- Outliers in the data expand the quantiles
- Skewed data might require different *k* to detect upper and lower outliers
- One-dimensional



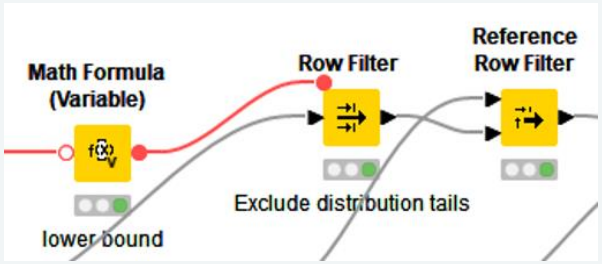
Flag data points outside the upper and lower whiskers of a box plot as outliers

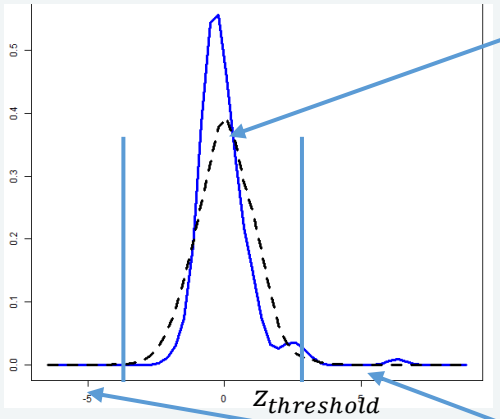


Z-score Based Outlier

Challenges:

- Normality assumption
- The parameters of the distribution are sensitive to outliers
- Doesn't work for data with a trend and seasonality





Standard Normal Distribution

Flag data points in the distribution tails as outliers

Outliers

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- Data Summary and Visualization
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  - Visualization for Higher Dimensions
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  - **Data Type (Numerical and Categorical) Transformation**
  - Data Normalization
  - String REGEX
- Feature Selection and Dimensionality Reduction
- Feature Engineering
- Data Integration
- Lab: Data Import, Filtering and Visualization
- Lab: Data Cleaning
- Review Questions



From Categorical to Numerical

- Binary attribute: numerical attribute with the values 0 and 1.
- Ordinal attribute ("sortable"): enumerate in the correct order $1, \ldots, k$
- Categorical attribute (not ordinal) with more than two values, say $a_1, \ldots, a_k$, should **not be converted into a single numerical attribute instead: convert to k attributes $A_1, \ldots, A_k$ with values 0 and 1.**
- a_i is represented by $a_i = 1$ and $a_j = 0$ for $i \neq j$ (1−of−n encoding).



From Numerical to Categorical

Splitting a numerical range into a number of bins

- **Equi-width discretization:** splits the range into intervals (bins) of the same length.
- **Equi-frequency discretization: s**plits the range into intervals such that each interval (bin) contains (roughly) the same number of records.
- **V-optimal discretization:** minimizes $\sum_i n_i V_i$ where n_i is the number of data objects in the i-th interval and V_i is the sample variance of the data in this interval.

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Normalization: Motivation



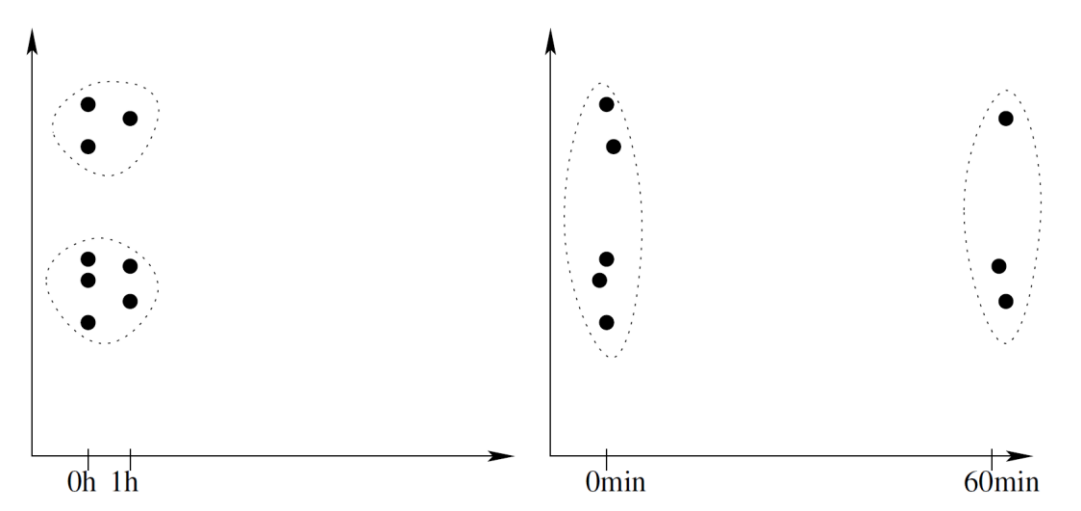
- For some data analysis techniques (e.g. PCA, MDS; cluster analysis) the influence of an attribute depends on the scale or measurement unit.
- To guarantee impartiality, some kind of **standardization** or **normalization** should be applied.

Example:

- Lengths in cm (100 – 200) and weights in kilogram (30 – 150) fall both in approximately the same scale
- What about lengths in m (1-2) and weights also in gram (30000 – 150000)?
  → The weight values in mg dominate over the length values for the similarity of records!

Normalization: Motivation

- 0h vs 1h can be expressed as 0min vs 60min



Goal of normalization:

- Transformation of attributes to make record ranges **comparable**

Construct Data: Assure Impartiality



- In absence of domain knowledge, different techniques can be applied
- min–max normalization $x \mapsto \frac{x-minX}{max_X-minX} : \mathbb{R} \to [0,1]$
- **z-score standardization** $x \mapsto \frac{x-\hat{\mu}_X}{\hat{\sigma}_X} : \mathbb{R} \to \mathbb{R}$
- **robust z-score standardization** $x \mapsto \frac{x-med}{IQR_X} : \mathbb{R} \to \mathbb{R}$

Robust to outliers!

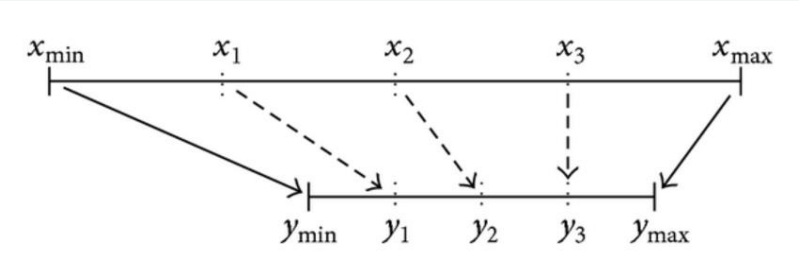
- **decimal scaling** $x \mapsto \frac{x}{10^s} : \mathbb{R} \to [0,1]$

Normalization: Techniques

- **min-max normalization**

$n: dom(X) \rightarrow [0,1]$

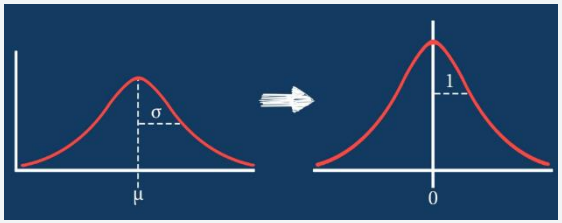
$$y = \frac{x - x_{min}}{x_{max} - x_{min}}$$



- **z-score normalization**

$s: dom(X) \rightarrow \mathbb{R}$

$$y = \frac{x - \hat{\mu}(X)}{\hat{\sigma}(X)}$$



- **normalization by decimal scaling**

$d: dom(X) \rightarrow [0,1]$

$y = \frac{x}{10^j}$ where j is the smallest integer value larger than $\log_{10}(\max(X))$

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Unstructured String Data



Data in string format is difficult to process (see unstructured data)

We can extract some information from string if the string feature has some pattern. For example the data below contains several useful information if properly cleaned

Job Title	Salary Estimate	Job Description	Rating	Company Name	Location	Headquarters	Size	Founded	Type of ownership	Sector	F
Senior Data Scientist	$111K-$181K (Glassdoor est.)	ABOUT	3.5	Hopper	New York, NY	Montreal, Canac	501 to 1000 employees	2007	Company - Private	Travel & Tourisn	U
Data Scientist, Product Analytics	$111K-$181K (Glassdoor est.)	At Noom, we	4.5	Noom US	New York, NY	New York, NY	1001 to 5000 employees	2008	Company - Private	Consumer Servic	U
Data Science Manager	$111K-$181K (Glassdoor est.)	Decode_M	-1	Decode_M	New York, NY	New York, NY	1 to 50 employees	-1	Unknown	-1	U
Data Analyst	$111K-$181K (Glassdoor est.)	Sapphire	3.4	Sapphire Digital	Lyndhurst, NJ	Lyndhurst, NJ	201 to 500 employees	2019	Company - Private	Information Tec	U
Director, Data Science	$111K-$181K (Glassdoor est.)	Director, Data	3.4	United	New York, NY	New York, NY	51 to 200 employees	2007	Company - Private	Business Service	U
Data Scientist	$111K-$181K (Glassdoor est.)	Job Brief	2.9	IFG Companies	New York, NY	Hartford, CT	201 to 500 employees	1985	Company - Private	Insurance	U
Quantitative Researcher	$111K-$181K (Glassdoor est.)	Experience:	4.4	PDT Partners	New York, NY	New York, NY	51 to 200 employees	1993	Company - Private	Finance	U
Quantitative Research Associate	$111K-$181K (Glassdoor est.)	Seeking a	-1	Enlightenment Resea	New York, NY	New York, NY	1 to 50 employees	-1	Unknown	-1	U
AI Scientist	$111K-$181K (Glassdoor est.)	Paige is a	5	Paige	New York, NY	New York, NY	1 to 50 employees	2018	Company - Private	Information Tec	U
Quantitative Researcher	$111K-$181K (Glassdoor est.)	About the	4.8	Jane Street	New York, NY	New York, NY	501 to 1000 employees	2000	Company - Private	Finance	U
Data Scientist	$111K-$181K (Glassdoor est.)	Company	3.9	Quartet Health	New York, NY	New York, NY	201 to 500 employees	2014	Company - Private	Information Tec	U
Data Scientist/Machine Learning	$111K-$181K (Glassdoor est.)	PulsePoint,Ñ¢,	4.4	PulsePoint	New York, NY	New York, NY	51 to 200 employees	2011	Company - Private	Information Tec	$
Data Scientist, Acorn AI Labs	$111K-$181K (Glassdoor est.)	Medidata:	4.3	Medidata Solutions	New York, NY	New York, NY	1001 to 5000 employees	1999	Company - Public	Information Tec	$
Data Scientist	$111K-$181K (Glassdoor est.)	A Career with	3.9	Point72	New York, NY	Stamford, CT	1001 to 5000 employees	2014	Company - Private	Finance	U
Data Scientist - Alpha Insights	$111K-$181K (Glassdoor est.)	Two Sigma is a	4.4	Two Sigma	New York, NY	New York, NY	1001 to 5000 employees	2001	Company - Private	Finance	U
Data Scientist	$111K-$181K (Glassdoor est.)	Data Scientist	3	Affinity Solutions	New York, NY	New York, NY	51 to 200 employees	1998	Company - Private	Business Service	U
Data Scientist, Analytics	$111K-$181K (Glassdoor est.)	Company Descri	3.6	Etsy	Brooklyn, NY	Brooklyn, NY	501 to 1000 employees	2005	Company - Public	Retail	$
Data Scientist/ML Engineer	$111K-$181K (Glassdoor est.)	Data	3.3	PA Consulting	New York, NY	London, United I	1001 to 5000 employees	1943	Company - Private	Business Service	$
Data Scientist	$111K-$181K (Glassdoor est.)	Job Description	3.6	Etsy	New York, NY	Brooklyn, NY	501 to 1000 employees	2005	Company - Public	Retail	$
VP, Data Science	$111K-$181K (Glassdoor est.)	We are looking	3.9	7Park Data	New York, NY	New York, NY	51 to 200 employees	2012	Company - Private	Business Service	U
Data Scientist, Disney+ Personaliz	$111K-$181K (Glassdoor est.)	Job Summary:Cc	4	Walt Disney Co.	New York, NY	Burbank, CA	10000+ employees	1923	Company - Public	Media	$
Senior Data Scientist, Data Scienc	$111K-$181K (Glassdoor est.)	We,Äôre	3.4	Squarespace	New York, NY	New York, NY	1001 to 5000 employees	2003	Company - Private	Information Tec	U
Quantitative Researcher ,Äì Intern	$111K-$181K (Glassdoor est.)	Job Description	4.1	Citadel Securities	New York, NY	Chicago, IL	201 to 500 employees	2002	Company - Private	Finance	U
Senior Data Engineer (Healthcare	$111K-$181K (Glassdoor est.)	Key	3.4	Enterprise	New York, NY	Jacksonville, FL	51 to 200 employees	1998	Company - Private	Information Tec	$
Data Scientist	$111K-$181K (Glassdoor est.)	Job Description	4.4	WITHIN	New York, NY	New York, NY	51 to 200 employees	2015	Company - Private	Business Service	U
Data Scientist, Marketplace Econo	$111K-$181K (Glassdoor est.)	We are looking f	3.8	Spotify	New York, NY	Stockholm, Swe	1001 to 5000 employees	2006	Company - Public	Information Tec	U
Data Scientist	$111K-$181K (Glassdoor est.)	About Datadog:\	4.1	Datadog	New York, NY	New York, NY	1001 to 5000 employees	2010	Company - Public	Information Tec	$
Lead Data Scientist	$111K-$181K (Glassdoor est.)	Description: Its	3.3	Aetna	New York, NY	Hartford, CT	10000+ employees	1853	Company - Public	Insurance	$
Data Scientist, Personalization	$111K-$181K (Glassdoor est.)	About	5	Hungryroot	New York, NY	New York, NY	1 to 50 employees	2015	Company - Private	Consumer Servic	$
Principal Data Scientist	$111K-$181K (Glassdoor est.)	Description: Its	3.3	Aetna	New York, NY	Hartford, CT	10000+ employees	1853	Company - Public	Insurance	$
Data Scientist	$120K-$140K (Glassdoor est.)	Caserta is a best	4.3	Caserta	New York, NY	New York, NY	51 to 200 employees	2001	Company - Private	Information Tec	U
Data Scientist, Decisions	$120K-$140K (Glassdoor est.)	At Lyft, our	3.7	Lyft	New York, NY	San Francisco, C	5001 to 10000 employees	2012	Company - Public	Information Tec	U

Regex

One way to clean the string data is through data **deletion** and **replacement** through the use of Regular Expressions (regex)

Regular expression is a pattern defining a class of strings. Some examples:

Given a column of strings

- "AnyWord" search for pattern "AnyWord"
- "word1|word2" search for values containing either "word1" or "word2"
- "Any.*Word" search for values containing Any and Word and anything inbetween the two words.
- "//$|//*" search for special character "$" or special character "*"
- "[a-zA-Z]" search for values containing any non numeric character
- "[a-zA-Z]{3,}" search for values containing at least 3 non-numeric character
- "[0-9]" search for values containing any numeric character

Regex in KNIME



The following nodes have Regex compatibility to transform string data:





Example String Manipulation in KNIME



Dialog - 4:2 - String Manipulation (Add Salary Range)

String Manipulation | Flow Variables | Memory Policy

Column List
ROWID
ROWINDEX
ROWCOUNT
Column0
index
Job Title
Salary Estimate
Job Description
Rating
Company Name
Location
Headquarters
Size
Founded
Type of ownership
Industry
Sector
Revenue
Competitors
Easy Apply

Flow Variable List
knime.workspace

Category: All

Function:
regexMatcher(str, regex)
regexReplace(str, regex, replaceStr)
removeChars(str)
removeChars(str, chars)
removeDiacritic(str)
removeDuplicates(str)
replace(str, search, replace)
replace(str, search, replace, modifiers)
replaceChars(str, chars, replace)
replaceChars(str, chars, replace, modifiers)
replaceUmlauts(str, omitE)
reverse(str)
string(x)

Description:
Applies regex to string and replaces str if regex matches.

Examples:
regexReplace("abc", "[a-zA-Z]{3}", "cba") = "cba"
regexReplace("aBc", "[a-zA-Z]{3}", "AbC") = "AbC"
regexReplace("abcd", "[a-zA-Z]{3}", "ABC") = "ABCd"

Expression:
```
1 regexReplace($Salary Estimate$, "[a-zA-Z]|\\$|\\(|\\)|\\.","")
2
```

Append Column: salRange
Replace Column: Easy Apply
Insert Missing As Null
Syntax check on close

OK | Apply | Cancel

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Is there such a thing as “too much data”?



"Too much data":

- Consumes storage **space**
- Eats up processing **time**
- Is difficult to **visualize**
- Inhibits ML algorithm **performance**
- Beware of the model: **Garbage** in ➔ Garbage out

Feature Selection vs. Dimensionality Reduction



Both methods are used for reducing the number of features in a dataset. However:

- Feature selection is simply selecting and excluding given features **without changing** them.
- Dimensionality reduction **might transform** the features into a lower dimension.
- Dimensionality reduction is a broader concept that includes feature selection

Feature Selection



- **Feature Selection**: choose a subset of the features (attributes) that is **as small as** possible and sufficient for the data analysis (= still informative!)

- Feature Selection includes:
  - Removing (more or less) **irrelevant features/fields** and
  - Removing **redundant features**

- **Evaluation** function to compare sets of attributes
- Strategy (heuristic) to select the possible feature subsets to be compared against each other with this measure

Feature Selection Techniques



- **Forward selection**

  Start with the **empty** set of features and **add** features one by one. In each step, add the feature that yields the best improvement of the **performance**.

- **Backward elimination**

  Start with the **full** set of features and **remove** features one by one. In each step, remove the feature that results in the smallest decrease in **performance**.

Feature Selection (Dimensionality Reduction) Techniques



The following are rather common example of feature selection (or dimensionality reduction). Data features are excluded based on:

- Ratio of missing values
- Low variance
- High Correlation

Dimensionality Reduction Based on Missing Values Ratio

First partition (as defined in dialog) - 0:337:0:276 - Partitioning (80% vs. 20%)

File Hilite Navigation View

Table "default" - Rows: 40000 | Spec - Columns: 231 | Properties | Flow Variables

Row ID	Var16	Var17	Var18	Var19	Var20	Var21	Var22	Var23	Var24	Var25	Var26	Var27	Var28
Row0	?	?	?	?	?	464	580	?	14	128	?	?	166.56
Row1	?	?	?	?	?	168	210	?	2	24	?	?	353.52
Row2	?	?	?	?	?	1212	1515	?	26	816	?	?	220.08
Row4	?	?	?	?	?	64	80	?	4	64	?	?	200
Row7	?	?	?	?	?	32	40	?	2	16	?	?	230.56
Row8	?	?	?	?	?	200	250	?	2	64	?	?	300.32
Row10	?	?	?	?	?	92	115	?	6	112	?	?	133.12
Row11	?	?	?	?	?	236	295	?	8	40	?	?	133.12
Row12	?	?	?	?	?	0	0	?	?	0	?	?	240.56
Row13	?	?	?	?	?	480	600	?	10	216	?	?	176.56
Row14	?	?	?	?	?	148	185	?	0	8	?	?	236.08
Row16	?	?	?	?	?	584	730	?	6	320	?	?	220.08
Row17	?	?	?	?	?	168	210	?	2	32	?	?	166.56
Row18	?	?	?	?	?	12	15	?	2	0	?	?	253.52
Row20	?	?	?	?	?	168	210	?	2	56	?	?	272.08
Row21	?	?	?	?	?	20	25	?	2	0	?	?	86.96
Row22	?	?	?	?	?	[illegible]	[illegible]	?	?	[illegible]	?	?	166.56
Row23	?	?	?	?	?	[illegible]	[illegible]	?	?	[illegible]	?	?	198.88
Row24	?	?	?	?	?	216	270	?	8	128	?	?	200
Row25	?	?	?	?	?	152	190	?	4	16	?	?	20.08
Row26	?	0	0	0	?	?	?	?	?	?	?	?	?
Row28	?	?	?	?	?	0	0	?	?	0	?	?	257.28
Row29	?	?	?	?	?	312	390	?	0	120	?	?	200
Row30	?	?	?	?	?	112	140	?	4	56	?	?	166.56
Row31	?	?	?	?	?	28	35	?	0	16	?	?	[illegible]
Row33	?	?	?	?	?	160	200	?	4	40	?	?	[illegible]
Row36	?	?	?	?	?	612	765	?	14	360	?	?	200
Row37	?	?	?	?	?	380	475	?	4	208	?	?	336.56
Row38	?	?	?	?	?	76	95	?	0	16	?	?	213.36
Row40	?	?	?	?	?	228	285	?	22	56	?	?	200
Row41	?	?	?	?	?	120	150	?	10	80	?	?	133.12
Row42	?	5	0	0	?	?	?	?	?	?	?	?	?
Row43	?	?	?	?	?	72	90	?	0	40	?	?	191.36
Row44	?	?	?	?	?	0	0	?	?	0	?	?	120.4
Row47	?	?	?	?	?	0	0	?	?	0	?	?	186.64
Row48	?	?	?	?	?	172	215	?	4	200	?	?	137.68
Row49	?	?	?	?	?	0	0	?	?	0	?	?	279.16

IF (% missing value > threshold) THEN remove column

Missing Value

Dimensionality Reduction Based on Low Variance



Output table - 0:347:0:337 - Missing Value (Numeric: 0)

File Hilite Navigation View

Table "default" - Rows: 40000 | Spec - Columns: 231 | Properties | Flow Variables

Row ID	20	Var21	Var22	Var23	Var24	Var25	Var26	Var27	Var28
Row51	h	336	420	0	8	72	0	0	133.12
Row52	h	120	150	0	0	16	0	0	286.96
Row54	h	124	155	0	0	0	0	0	234.72
Row55	h	184	230	0	4	64	0	0	642.64
Row56	h	268	335	0	4	88	0	0	133.12
Row57	h	128	160	0	0	96	0	0	198.88
Row59	h	132	165	0	0	112	0	0	253.52
Row60	h	44	55	0	0	24	0	0	186.64
Row61	h	104	130	0	4	72	0	0	166.56
Row62	h	212	265	0	6	136	0	0	379.6
Row63	h	20	25	0	0	0	0	0	166.56
Row65	h	492	615	0	18	256	0	0	133.12
Row66	h	148	185	0	2	8	0	0	186.64
Row68	h	140	175	0	2	40	0	0	176.56
Row69	h	0	0	0	0	0	0	0	166.56
Row71	h	0	0	0	0	0	0	0	392.08
Row72	h	124	155	0	6	88	0	0	153.2
Row73	h	152	190	0	0	32	0	0	253.52
Row74	h	324	405	0	8	104	0	0	186.64
Row75	h	0	0	0	0	0	0	0	0
Row76	h	60	75	0	6	0	0	0	200
Row77	h	180	225	0	4	88	0	0	166.56
Row78	h	232	290	0	4	144	0	0	200
Row79	h	16	20	0	0	16	0	0	313.68
Row81	h	152	190	0	0	48	0	0	220.08
Row82	h	108	135	0	4	88	0	0	166.56

Note: requires min-max-normalization, and only works for **numeric** columns

If column has **constant** value (variance = **0**), it contains no useful information

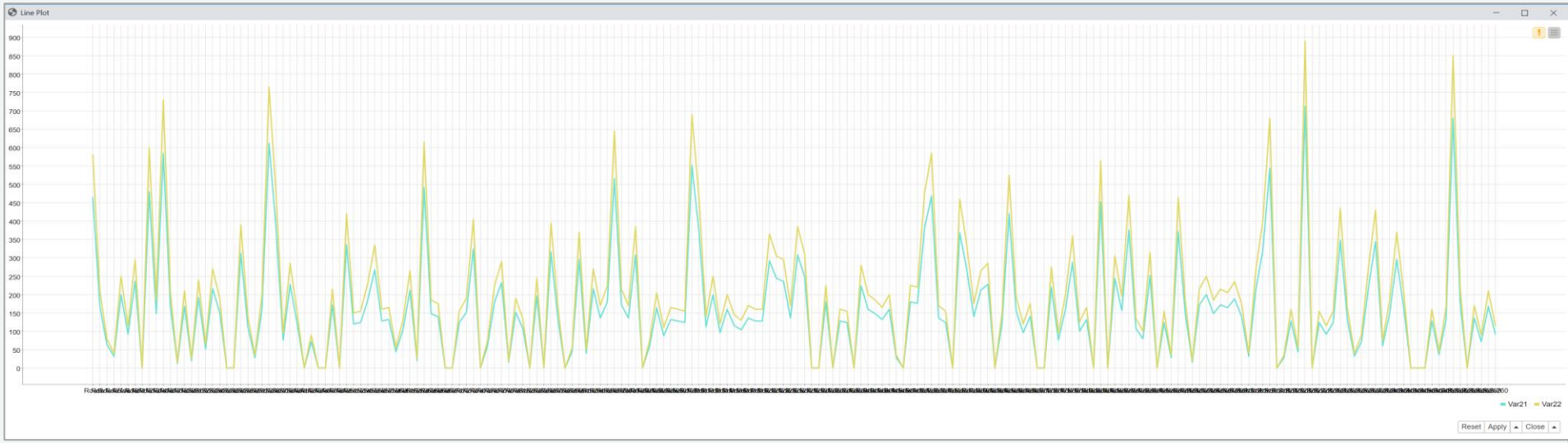
In general: **IF (variance < threshold) THEN remove column**

Dimensionality Reduction Based on High Correlation

Two **highly correlated** input variables probably carry similar information

$$IF\ (corr(var1, var2) > threshold\ \Rightarrow remove\ var1$$

Note: requires min-max-normalization of numeric columns



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What is Feature Engineering?



- **Feature engineering** is a machine learning technique that leverages data to create new variables that aren't in the dataset.

- Example: prices of properties in some city

new attribute

Sq Ft.	Amount
2400	9 Million
3200	15 Million
2500	10 Million
2100	1.5 Million
2500	8.9 Million

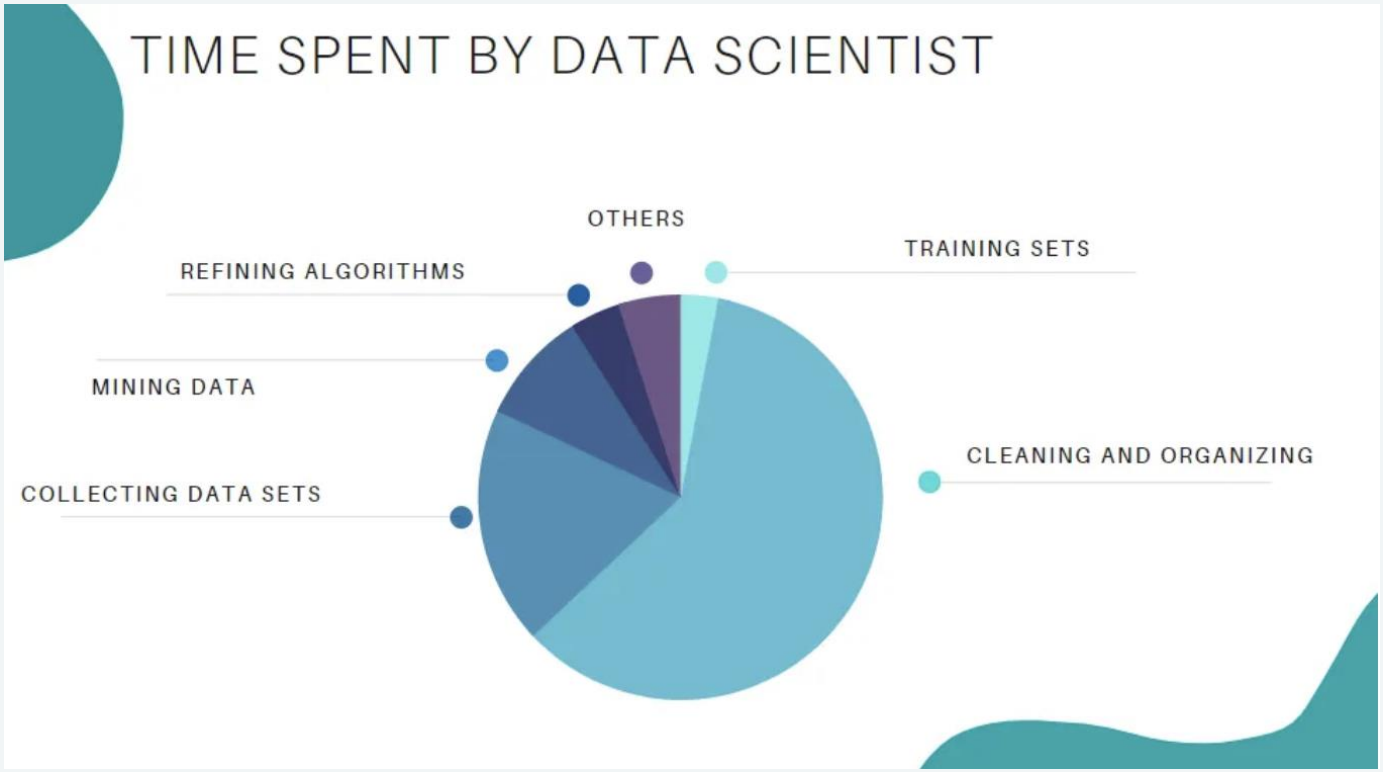
Original dataset

Sq Ft.	Amount	Cost Per Sq Ft
2400	9 Million	4150
3200	15 Million	4944
2500	10 Million	3950
2100	1.5 Million	510
2500	8.9 Million	3600

New dataset

Why we do need Feature Engineering?

- Data scientists spend most of their time to clean data

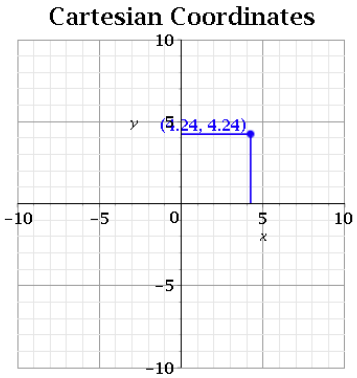


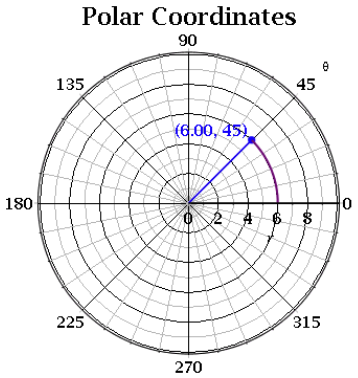
- Feature engineering can **simplify** and **speed up** data transformations while also **enhancing model accuracy**

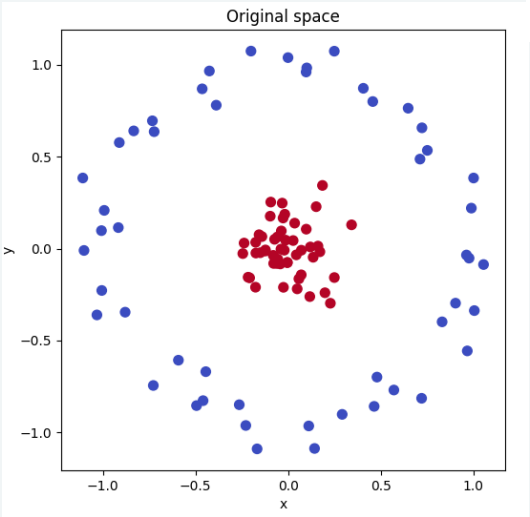
Feature Engineering Example

Sometimes transforming the original data leads to better modeling results

Euclidean to polar coor

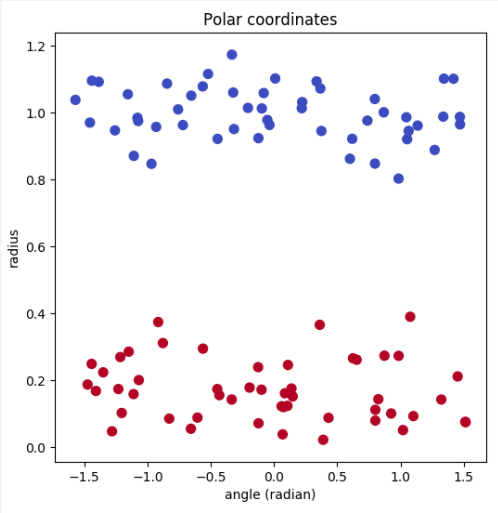






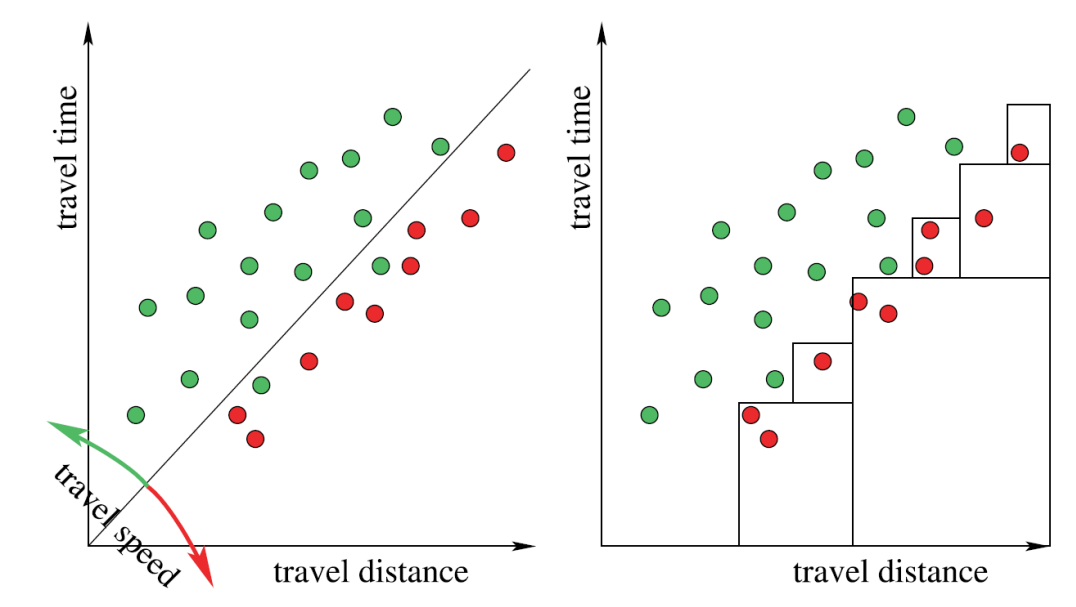
Radius $r = \sqrt{x^2 + y^2}$

Angle $\theta = Tan^{-1}(y/x)$



Feature Engineering Example

Offering new features that were difficult to represent by the original model



Example for the usefulness of derived features:
for a number of journeys, the travel time and distance are shown; the color indicates whether the driver was ticketed or not. Discriminating both classes with axis-parallel rectangles is laborious, but easy with a new attribute for travel speed

Feature Engineering in practice



Feature engineering includes all **transformation techniques** of existing attributes and and **construction** of new attributes that may (or may not) replace the original attributes

- Exploit domain knowledge to improve the model results

Example: Feature Engineering by Scale Conversion



Scale Conversion

- Categorical → Numerical: map categorical and ordinal values to a set of binary values
- Numerical → Categorical: **Discretization** (equal-width, equal-depth, V-optimal)

Example: Engineering Ratio Attributes



Example: **Find the best workers in a company.**

Attributes available:

- the tasks, a worker has finished within each month,
- the number of hours he has worked each month,
- the number of hours that are normally needed to finish each task.

These attributes do *contain* information about the efficiency of the worker. But instead of using these three "raw" attributes, it might be more useful to define a new attribute ***efficiency***.

$$efficiency = \frac{hours\ actually\ spent\ to\ finish\ the\ tasks}{hours\ usually\ spent\ to\ finish\ the\ tasks}$$

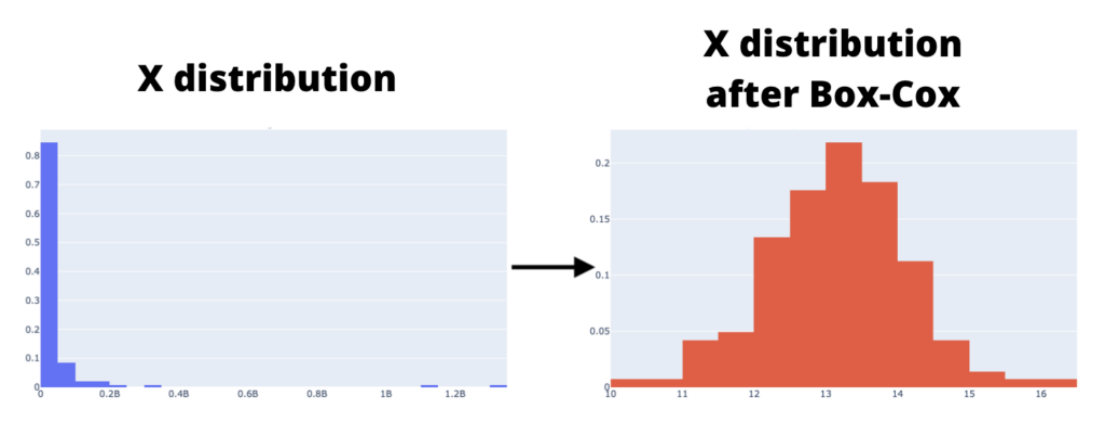
Example: Feature Engineering with Box-Cox Transform

Typical assumption: some variables obey a certain distribution (e.g. Gaussian)

- Transform the data to better approximate the distribution using the **power transform (Box-Cox Transform)**

$$y \mapsto \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda \overline{y}^{(\lambda-1)}} & if\ \lambda \neq 0 \\ \overline{y} \log y & if\ \lambda = 0 \end{cases}$$

Note: Only idea behind those techniques



Complex Data Types and Feature Extraction



Complex data types:

- Texts
- Graphs
- Images
- Molecules
- Other Objects

Especially for complex data types, **feature extraction is required**

- **Text data analysis.** Frequency of keyword, . . .
- **Time series data analysis.** Fourier or wavelet coefficients, . . .
- **Image data analysis.** Fourier or wavelet coefficients, . . .
- **Graph data analysis.** Number of vertices, number of edges, . . .

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  - Visualization for 1 or 2 Dimensions
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  - Missing Values Imputation
  - Outliers
  - Data Type (Numerical and Categorical) Transformation
  - Data Normalization
  - String REGEX
- Feature Selection and Dimensionality Reduction
- Feature Engineering
- **Data Integration**
- Lab: Data Import, Filtering and Visualization
- Lab: Data Cleaning
- Review Questions

Data Integration



Vertical Data Integration (Concatenation)

- Unify database structures
- Remove duplicates

id	Last name	First name	Gender
p2	Mayer	Susan	F
p5	Smith	Walter	M
p7	Brown	Jane	F
…	…	…	…

+

Shopper id	Item id	Price
p2	i254	12.50
p5	i4245	1.99
p5	i32123	1.29
p5	i254	12.50
p5	i21435	5.99
p7	i254	12.50
…	…	…

Horizontal Data Integration (Join)

- Overrepresentation of items
- Data explosion

Item id	Price	Last name	First name	Gender
i254	12.50	Mayer	Susan	F
i4245	1.99	Smith	Walter	M
i32123	1.29	Smith	Walter	M
i254	12.50	Smith	Walter	M
i21435	5.99	Smith	Walter	M
i254	12.50	Brown	Jane	F
…	…	…	…	…

The two data sets on top contain information about customers and product purchases. The joint data set at the bottom combines these two tables. Note how we loose information about individual customers and how a lot of duplicate information is introduced. In reality this effect is, of course, far more dramatic

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Lab: Visualization of sales data

Data: sales_data.csv

Steps:

1. Read csv data file
2. Filter columns and rows
3. Visualize your results in different charts. (2 charts)

Step 1: Create a new KNIME workflow



- Download sales_data.csv
- Create a new workflow (ch2_lab_YOUR_ID)
- Drag the file to the workflow to import the data

Step 2: Data Filtering



Study the data table and decide upon filtering operation

- Add a Column Filter node. Configure to exclude a selected column
- Add a Row Filter note. Configure to exclude rows by attribute value for unknow for all columns (amount, country, date).

Step 3: Visualization



- Add a Stacked Area Chart Node and Pie/Donut Chart Node. Configure the two charts (Think about the information to display, what gives more insights)

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Data Cleaning



Import adult.csv and clean the data.

Your workflow must contain at least 3 nodes of the following. Each node operation must make sense from a data cleaning point of view

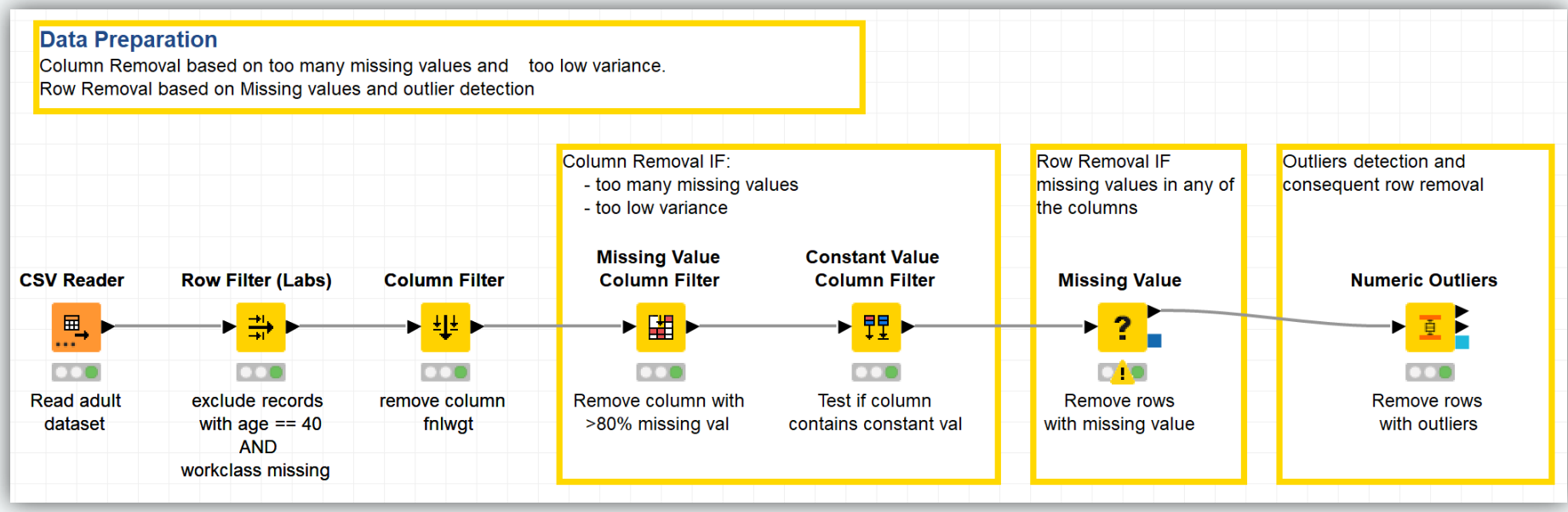
Possible operations:

- Row Filter
- Column Filter
- Missing Value Filter
- Outliers Detection
- Data Normalization: check nodes Normalizer, Normalizer (Apply) and Denormalizer
- Data Transformation: nodes Rule Engine, Math Formula, String Manipulation

In each node explain what you did.

Example Workflow

This workflow applies simple filtering operations



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Review Questions

Question 1: Explain what outlier is. How to handle the outlier if it is found?

Question 2: Given the data field

X: [50, 23, 13, 4, 9, 3, 19, 29, 53, 11]

Please find out the followings:

- 1st quartile
- 2nd quartile
- 3rd quartile
- Interquartile range (IQR)

Question 3 (15 points): Given the following data:

Age = [10 15 15 18 19 20].

Calculate the **min-max** normalized values for Age

Review Questions - Multiple Choices

1. Box plot can visualize the following characteristics of data, except for …

   (A) Median (B) Interquartile Range (IQR) (C) Trend (D) Outliers

2. Which of the following is not a dimension reduction technique?

   (A) Scatter plot works well with very large data sets

   (B) Scatter plot can help identify outliers

   (C) Sunburst chart can display multidimensional hierarchical data in a radial layout

   (D) Radar plot and parallel plot are essential similar, but laying out axes differently

3. What is the **median** value of (1, 2, 3, 4, 5, 10, 20, 30)?

   (A) 4 (B) 5 (C) 4.5 (D) 9.375 (E) 10

4. What is the mean value of (10, 50, 30, -10, 20, 20)?

   (A) 10 (B) 0 (C) 50 (D) 20 (E) None of the mentioned

Review Questions - Multiple Choices

5. Which of the following is not a dimension reduction technique?

(A) Ratio of missing values (B) Low variance (C) High correlation (D) Cross-Validation

6. Which of the following is not a normalization technique?

(A) Min-Max normalization (B) Z-score standardization (C) Decimal scaling (D) Divide by Min (E) Robust Z-scores standardization (IQR)

7. Which of the following is the most common approach to impute missing values?

(A) The median value for all records (B) The max value for all records (C) The min value for all records (D) a value of 0 (E) a value of 100

Review Questions - Multiple Choices

8. In a database a value for “weight” is missing. Of the following reasons one is particularly severe as it can highly impact the analysis if the record is deleted. Which reason is it?

(A) the value is missing completely at random

(B) the value is missing because the respondent did not know her weight

(C) the value is missing because the respondent did not want to reveal that he is obese

(D) the value is missing because the data got accidentally deleted

(E) the value is missing because the respondent was not sure she should enter kg or pounds

9. When a KNIME workflow does not give the expected results, you may try to fix it by the following measures, except for …

(A) Check if there is any warning/error message in the console. If yes, look into and resolve it

(B) Email the instructor/TA and ask for help

(C) Check if each node has been configured properly or not